PYTHON DATA SCIENCE · FINAL ASSESSMENT

Capstone: The Pinecrest Report

The single end-of-course assessment (Lesson 8.1). Each learner chooses one of four business questions about Pinecrest Outfitters, answers it from the course data in a notebook that reruns from a fresh start, and reports in one page with one or two figures. Score it with the lesson's six-row rubric below. This page also holds the instructor answer notes for all four questions: it is for instructors only and nothing on it goes into a learner-facing file.

Covers **Modules 1–7** | Started **Session 20**, studio **Session 21** | Submit within **1 week** | Successful: **3+** on every row | Instructor only

THE BRIEF (AS THE INSTRUCTOR SEES IT)

From Lesson 8.1, the memo from Pinecrest leadership: "We have four questions that keep coming up in meetings, and opinions on all of them. We would like evidence. Please pick one, investigate it with the data we have, and send us one page: what you found, how sure you are, and what you could not tell. We will read the page, not the code, but someone on our side may rerun your notebook to check the numbers."

QUESTION	MAIN DATA
A · Why is Asheville falling behind?	`daily_sales.csv`; possibly `store_weather.csv`, `survey.csv`, `sales_2025.xlsx`
B · Does it matter how we found the customer?	`customers.csv`, `orders.csv`
C · Does the welcome discount bring people back?	`orders.csv`; possibly `customers.csv`
D · What did the March 2025 price increase do to sales?	`order_items.csv`, `orders.csv`, `products.csv`

A learner may propose their own question if it names a decision, the data exist in `data/`, and no lesson project already answered it (online footwear returns, the checkout A/B test, staffing and revenue, rain and visitors are taken). Approve it in writing before Session 21.

- **Deliverables:** `pinecrest_report.ipynb` (runs from a fresh start and produces every number and figure), one or two figures (PNG, finding-first titles), `report.md` (one page: Question, Answer, Evidence, Caveats, Next questions, Method), `README.md`; optional: a Streamlit dashboard or a Plotly or Bokeh file.
- **Time (420 minutes in the lesson):** before Session 21, Milestones 1 to 4 (195 minutes, started in the Session 20 kickoff); in the Session 21 studio, Milestone 5 and most of 6 (90); after class, the rest of 6, Milestone 7 and Milestone 8 (75); the optional dashboard (60).
- **Assessed:** the submitted folder, against the six rubric rows. The Session 21 showcase is practice and is not scored; note one strength per learner from it for the written feedback.
- **There is no answer key for learners.** The notes below are for coaching and grading only. An honest, well-quantified "no clear difference" scores exactly as well as a discovery: two of the four questions (B and C) have one. Questions A and D have clear findings, and the notes for D say what an over-claim of a real effect looks like.

THE RUBRIC (LESSON 8.1, REPRODUCED EXACTLY)

CRITERION	1 · BEGINNING	2 · DEVELOPING	3 · PROFICIENT	4 · EXEMPLARY
Question weight 15%	A topic, not a question ("look at Asheville"). No plan.	A question is stated, but the measures and the comparison are decided as the code goes along.	The question, the decision it informs, the measures and the comparison are written down before the analysis, and the work stays on them.	Also states in advance what result would change the reader's mind, and revises the plan openly when the data requires it.
Data handling weight 20%	Files loaded and used as they are. Row counts never checked.	Some checks (shape, missing values), but merges and filters are not verified, or rows vanish without comment.	Every load, merge and filter is followed by a check of row counts or keys. Missing values, unmatched keys and odd values are investigated and each decision is stated.	Also shows that the conclusion holds under a different reasonable choice (for example with and without outliers, or another definition).

CRITERION	1 · BEGINNING	2 · DEVELOPING	3 · PROFICIENT	4 · EXEMPLARY
Analysis correctness weight 25%	Raw averages or totals compared with no measure of uncertainty.	A test is run, but its assumptions do not fit the data, the comparison is unfair, or only a p-value is reported.	Like is compared with like. The method suits the data type and shape. Each result has an estimate, a confidence interval, a test and an effect size.	Also addresses multiple comparisons or non-independence where relevant, and considers an alternative explanation and tests it.
Visuals weight 10%	Default charts with unlabeled axes, or many charts with no message.	Readable charts with labels, but titles name the topic, or the chart type hides the comparison.	One or two figures, each with a title that states the finding, labeled axes with units, purposeful color, error bars identified, saved from code at a stated size.	The figure alone carries the argument. A reader who sees nothing else understands the answer and its uncertainty.
Communication weight 20%	The report retells the steps taken. No clear answer.	An answer is given, but numbers lack uncertainty, caveats are generic ("more data is needed"), or it runs well past a page.	One page that leads with the answer, supports it with quantified evidence, gives specific caveats and next questions, in plain language.	Also never says more than the evidence allows: wording about cause is exact, and a non-specialist could act on it.
Reproducibility weight 10%	The notebook fails or gives different numbers when rerun.	It runs, but depends on cells run out of order, absolute paths, or unseeded randomness.	Restart and Run All succeeds from the project folder with relative paths and seeds. Every number in the report can be found in the notebook's output.	Also includes a short README, pinned requirements, and repeated logic in functions with clear names.

WEIGHTING AND THE RESULT

- Score each row 1 to 4 (whole numbers; no half points). The weighted score is the sum of score × weight, out of 4.00. Weights: Question 15%, Data handling 20%, Analysis correctness 25%, Visuals 10%, Communication 20%, Reproducibility 10%.
- **Successful capstone:** 3 (Proficient) or higher on every row, as the lesson states. Report the weighted score alongside it (3.00 to 4.00).
- **Successful with a revision:** one row at 2 and the weighted score at least 2.75. The learner fixes that row within one week; re-score that row only.
- **Not yet:** any row at 1, two or more rows at 2, or a notebook that does not run. Return written feedback (one thing done well in each row and the single change that would most improve the work, as in the lesson's peer-review challenge) and allow one resubmission within two weeks.
- **Reproducibility first:** run the notebook before reading anything else. If it fails, ask for a fix within 48 hours before scoring the other rows.

SCORING GUIDANCE BY ROW

ROW	WHAT TO LOOK FOR
Question	Read the first Markdown cell and the report's Question line. Proficient needs the decision, the measures with definitions and the main comparison written before the analysis; look for a plan that the notebook then follows (or says where it changed and why).
Data handling	Look for row counts or key checks after every load, merge and filter, and a sentence for each decision about missing, unmatched or odd rows. A merge without a printed count is Developing at best. Exemplary shows the conclusion under a second reasonable choice.
Analysis correctness	The heaviest row. Check the comparison is fair for the question chosen (the traps below), the method fits the data (proportions, skewed amounts, several groups), and the main result has all four of estimate, 95% interval, test and effect size. A significant result from an unfair comparison is Developing even when the code is flawless.
Visuals	Judge the one or two figures the report uses, not the exploration. Title states the finding; axes and units; error bars named in a caption; saved by code. Exemplary: you can state the answer and its uncertainty from the figure alone.
Communication	Read report.md first, as the brief's reader would. Answer first, every number with its uncertainty, caveats specific to the data, about one page. Wording about cause must match the design. A well-reported null is scored exactly like a positive finding.

ROW	WHAT TO LOOK FOR
Reproducibility	Restart and Run All in a fresh kernel from the project folder with the course data/ beside it. Then pick three numbers in report.md and find them in the output. Any number you cannot find is a Developing mark here and a question for the learner.

SCORING SHEET

Learner: _____ · Question: A / B / C / D / own · Date received: _____ · On time: yes / no

ROW	WEIGHT	SCORE 1–4	SCORE × WEIGHT	EVIDENCE (CELL OR LINE IN REPORT.MD)
Question	15%			
Data handling	20%			
Analysis correctness	25%			
Visuals	10%			
Communication	20%			
Reproducibility	10%			
Weighted total	100%			Result: successful / successful with revision / not yet

SUBMISSION AND ACADEMIC HONESTY

- **Deadline:** one week after Session 21, as a zipped project folder (without the course `data/` folder) or a link to a repository. Every learner submits their own work, even after working alongside others.
- **AI assistants and code from the web are allowed, on the lesson's terms:** "Use the tools a working analyst would use, on the same terms: you are responsible for every line and every claim." The README must say what help was used (assistant, website, classmate) and for what. Undisclosed help that the learner cannot explain is treated as not their work.
- **The explain-two-cells check:** for any submission, you may pick two cells and ask the learner to explain, in five minutes, what they do and why the result is right. A learner who cannot explain a cell revises it; the row it affects is scored after the revision.
- **Discussion is encouraged; shared code is not.** Learners on the same question may compare definitions and traps; their notebooks, figures and wording must be their own.

- **Data:** the course data are fictional and belong to the course. The README says so ("Pinecrest Outfitters is a fictional company. The data was written for the Python Data Science course by Ray de la Paz.") and links to the data instead of republishing it. No real or work data, no secrets in any cell or output.
- **Do not share these notes** with learners before or after grading; feedback refers to the rubric and to the learner's own work.

LATE WORK AND ALTERNATIVES

- **Extension:** a learner may ask for one extra week before the deadline, no reason needed. Later requests are at your discretion.
- **Late without an extension:** still scored on the rubric; the result is recorded with the date received. Do not deduct points: the rubric measures the work, and the record shows the timing.
- **Missed Session 21:** the learner records a three-minute talk (screen and voice) about their one figure and submits it with the folder, and joins a 15-minute check-in with you within the week in place of the studio coaching.
- **Missed Sessions 20 and 21, or a narrower scope agreed in advance:** one question, one comparison, one figure. All six rows still apply; Question and Analysis correctness are judged on the narrower plan.
- **Accessibility:** the showcase can be a written gallery card instead of a talk; the report can be read aloud to you instead of submitted as text where that suits the learner. The analysis itself is the same.

RECORDING THE TWO CHECKPOINTS ALONGSIDE THE CAPSTONE

The course has two short checkpoints before the capstone: **Checkpoint 1 in Session 13** (an open-notebook analysis task on Modules 1 to 4, scored with its own five-row rubric in the Session 13 answer key) and **Checkpoint 2 in Session 18** (a data-access task on Modules 5 and 6, in the Session 18 answer key). They are formative: they tell you who needs support before the capstone and are not weighted into the capstone score.

LEARNER	CHECKPOINT 1 (S13) SCORE / STATUS	CHECKPOINT 2 (S18) STATUS	CAPSTONE QUESTION	CAPSTONE ROWS (Q, D, A, V, C, R)	WEIGHTED	RESULT

- **Record each checkpoint** as its rubric score and a status: Met (the checkpoint's own pass line) or Not yet. A learner may redo a Not yet checkpoint once, at home, before the capstone deadline.
- **Course completion** = both checkpoints Met (or redone) and a successful capstone. A learner whose capstone is successful but a checkpoint is Not yet completes when the checkpoint is redone.

- **Use the checkpoints for coaching:** a Not yet on Checkpoint 1 predicts trouble in Data handling and Analysis correctness; on Checkpoint 2, in loading and merging. Visit those learners first in the Session 21 studio.

ANSWER NOTES · QUESTION A: WHY IS ASHEVILLE FALLING BEHIND?

A SOUND ANALYSIS: TABLES AND STEPS

1. Use `daily_sales.csv` (register totals, walk-ins included): one row per store-day, 2,903 rows; 6 missing revenue values (one in Asheville, 2024-02-16); Asheville has 364 rows in 2024 and 363 in 2025.
2. Aggregate per store and year: visitors, transactions, revenue. Decompose revenue = visitors × conversion (transactions ÷ visitors) × basket (revenue ÷ transaction); on a log scale the three changes add up exactly.
3. Make the comparison fair in time: each month against the same month a year earlier, or each day against the same weekday 52 weeks earlier; and fair across stores: the same decomposition for Bend, Boise and Flagstaff.
4. Test the factor that moved (paired by matched day or by month) and show the factors that did not move with their intervals.
5. Optional context: `store_weather.csv` (did rain change?), `survey.csv` (a single October to November 2025 snapshot), `sales_2025.xlsx` Targets.

KEY NUMBERS (FROM THE INSTRUCTOR'S OWN RUN ON THE COURSE DATA)

- Revenue 2024 → 2025: Asheville $2,123,278 → $2,017,699 (**−5.0%**); Bend +3.8%, Boise +9.3%, Flagstaff +19.7%.
- Asheville visitors 103,676 → 95,915 (**−7.5%**; 285.6 → 264.2 per day). Conversion 23.27% → 23.71% (+1.9% relative). Basket $88.03 → $88.74 (+0.8%). In log points: revenue −5.10 = visitors −7.78 + conversion +1.87 + basket +0.81.
- The other three stores: conversion and basket change by under 1% each; their visitors move with their revenue (+3.1%, +9.2%, +20.7%). Asheville is unusual in traffic only.
- Visitors per day against the same month of 2024: down in **12 of 12 months** (−3.4% to −12.5%); mean monthly change −7.5% (95% CI −9.6% to −5.4%, one-sample t p = 8e-6). Against the same weekday 52 weeks earlier: **−21.6 visitors a day** (95% CI −26.9 to −16.4; paired t p = 6e-15; d = −0.43; n = 362 days).
- Daily conversion 23.29% → 23.65%: +0.36 points (95% CI −0.06 to +0.78; Welch p = 0.094; d = 0.12). Daily basket $88.19 → $88.70: +$0.51 (95% CI −$0.65 to +$1.67; p = 0.39; d = 0.06).
- Not the weather: rainy-day share 23.2% → 23.8%; visitors fell 7.6% on dry days and 6.3% on rainy days. Survey: Asheville staff satisfaction 3.62 against 4.02 to 4.16 elsewhere (a snapshot, not a trend). Account-holder orders in `orders.csv`: completed store revenue −13.3%, orders −6.8% (a different basis). 2025 revenue reached about 91% of target.

THE ANSWER THE DATA SUPPORT

A traffic problem: about 7.5% fewer people came through the door in 2025, in every month of the year, while the share who bought and what they spent stayed the same (both differences are small and their intervals include zero). The data say what kind of problem it is, not why: nothing in these files explains the lost visitors.

TRAPS

- **Reading revenue alone,** or concluding "customers spend less" without decomposing.
- **Season or calendar:** comparing different months or unmatched weekdays; partial-year comparisons.
- **Mean of daily ratios vs ratio of sums** used interchangeably without saying which (they differ slightly: 23.29% vs 23.27% for 2024).
- **The survey as a cause.** One snapshot cannot explain a two-year trend; low staff ratings are a lead for the next question, not a finding.
- **Basis:** `orders.csv` holds account holders only (Asheville −13.3% there vs −5.0% at the registers). Fine if stated, wrong if mixed.
- **Non-independence:** 363 days from one store are not 363 independent customers; month-level or matched-day comparisons are the honest unit.
- **Staffing:** staff fell 5.47 → 5.22 a day, but staff are scheduled for expected demand (the Lesson 4.4 confound), so this is a consequence, not a cause.

WHAT AN HONEST "NO CLEAR DIFFERENCE" SOUNDS LIKE HERE

For conversion and basket: "Conversion (+0.36 points, 95% CI −0.06 to +0.78) and spend per transaction (+$0.51, CI −$0.65 to +$1.67) did not change measurably; the decline is in visitors." A learner who reports "Asheville's conversion is fine" with those intervals has the honest-null standard.

FOLLOW-UP QUESTIONS TO ASK THE LEARNER

1. "Which factor moved, by how much in each month, and what do the other stores' same factors do?"
2. "What would you collect to find out why fewer people come in: door-counter checks, local competition, foot traffic in the street, store hours?"
3. "Is the drop larger in some months, and would you expect that if the cause were the weather?"

WHAT A WEAK SUBMISSION TYPICALLY GETS WRONG

- Reports −5% revenue and stops, or names a cause (staff, prices, weather) with no test.
- Compares Asheville 2025 with Boise 2025 (level differences between stores are not a decline).
- Uses the survey's staff rating as the explanation.

ANSWER NOTES · QUESTION B: DOES IT MATTER HOW WE FOUND THE CUSTOMER?

A SOUND ANALYSIS: TABLES AND STEPS

1. Build one row per customer from `customers.csv` (4,850) with completed-order count and spend from `orders.csv` merged left, `validate="one_to_one"`; customers with no completed order get 0 (1,112 of them).
2. Note the 739 orders of 150 closed accounts with no customer row: they have no channel and are left out, by a stated decision.
3. Define "worth" before looking: total completed spend over the window, number of orders, and whether they ever bought.
4. Make time fair: the data cover 2024 and 2025 only, and a third of customers signed up during that window. Either restrict to customers who signed up before 2024-01-01 (all have the full 24 months: 3,263) or use spend per month of exposure.
5. Five groups and skewed spend: Kruskal-Wallis with an effect size (epsilon squared), a chi-square for ever bought, and at most a few planned pairwise comparisons with Holm adjustment; bootstrap intervals for the headline difference.

KEY NUMBERS (FROM THE INSTRUCTOR'S OWN RUN ON THE COURSE DATA)

- All customers, mean completed spend: organic $709, paid search $727, social $706, referral $772, in store $699 (medians $408 to $463). Kruskal-Wallis H = 2.73, **p = 0.60**; ever bought 76% to 80% (chi-square p = 0.48, Cramér's V = 0.03).
- Signup years are spread alike across channels (66% to 68% of every channel signed up before 2024), so tenure does not bias this comparison, but it must be checked to know that.
- Full-window cohort (signed up before 2024, n = 3,263): mean spend $827 / $857 / $861 / $891 / $839; medians $552 to $622; Kruskal-Wallis H = 2.03, **p = 0.73**, epsilon squared ≈ 0; ever bought 80.5% to 84.6% (p = 0.25, V = 0.04).
- All 10 pairwise Mann-Whitney comparisons: smallest raw p = 0.166 (organic vs referral); every Holm-adjusted p = 1.0; rank-biserial correlations within ±0.05.
- Referral against all other channels (full-window cohort): **$891 vs $843, +$48 (95% bootstrap CI −$46 to +$146)**, ratio 1.06, d = 0.05, Welch p = 0.35. The upper end of the interval is about 1.17 times: "twice as much" is ruled out.
- Spend per month of exposure, all customers: $34.1 to $37.6 by channel (Kruskal-Wallis p = 0.62). Standard deviation of spend in the cohort: $925, so differences of $50 between channels are well inside the noise.

THE ANSWER THE DATA SUPPORT

No meaningful difference. Customers from all five channels spend about the same over two years and are about equally likely to buy at all; the referral advantage is about 6% and could be anywhere from 5% less to 17% more. The belief that referred customers are worth twice as much is not supported: the data rule it out.

TRAPS

- **Tenure:** comparing customers who joined in 2025 with those who joined in 2019 on two-year spend.

- **Dropping non-buyers** with an inner merge, which changes the question to "among buyers" without saying so.
- **Ten pairwise tests without adjustment,** then reporting the one with the smallest p.
- **Means of a skewed amount** with t-tests and no interval (maximum customer spend $7,946; 12 bulk-order customers spread across four channels).
- **Controlling for loyalty tier:** tier is defined by spend, the outcome itself.
- **"Worth"** defined after looking.

WHAT AN HONEST "NO CLEAR DIFFERENCE" SOUNDS LIKE HERE

"Referred customers spent $891 on average against $843 for the rest (95% CI for the difference −$46 to +$146; p = 0.35; d = 0.05). We found no evidence that the acquisition channel matters for spend, and the data rule out a referral advantage much above 17%."

FOLLOW-UP QUESTIONS TO ASK THE LEARNER

1. "Does every customer in your comparison have the same amount of time to spend?"
2. "How many comparisons did you look at before choosing the one in your report?"
3. "If the channels do not differ in value, what should Marketing compare instead: the cost of acquiring a customer in each channel?"

WHAT A WEAK SUBMISSION TYPICALLY GETS WRONG

- Reports the naive referral mean ($772 vs about $710) as "9% more valuable" with no interval.
- Runs ten t-tests and highlights one.
- Drops non-buyers silently, or keeps the 739 orphan orders under a made-up channel.

ANSWER NOTES · QUESTION C: DOES THE WELCOME DISCOUNT BRING PEOPLE BACK?

A SOUND ANALYSIS: TABLES AND STEPS

1. From `orders.csv`, sort by customer and date and take each customer's first order in the data (3,942 customers). WELCOME10 was used 1,139 times, always on that first order (28.9% of first orders) and never later.
2. Decide whose first order is a true first order: the data start on 2024-01-01, so 2,865 of the 3,942 customers signed up before 2024 (`customers.csv`); 150 first orders belong to closed accounts with no signup date. The cleanest group is customers who signed up in 2024 or 2025 (1,077, of whom 307 used the code).
3. Define "came back" with a fixed follow-up window (90, 180 or 365 days after the first order) and keep only customers whose first order was at least that long before 2025-12-31.

4. Compare two proportions: the difference with a 95% interval, a two-proportion z-test, Cohen's h. Optionally, repeat orders and spend in the year after the first order.
5. Say how large a difference the data could have detected, and put the cost of the code next to it.

KEY NUMBERS (FROM THE INSTRUCTOR'S OWN RUN ON THE COURSE DATA)

- **The naive comparison (the trap):** came back at all, any follow-up: 83.8% with the code vs 80.9% without, +2.8 points (95% CI +0.1 to +5.3), p = 0.039. It looks like a finding.
- All first orders, fixed windows: 90 days 47.6% vs 45.7% (+1.9 points, CI −1.6 to +5.5, p = 0.29, h = 0.04); **180 days 68.9% vs 67.1%** (+1.8, CI −1.6 to +5.3, p = 0.29); 365 days 86.6% vs 85.4% (+1.2, CI −1.5 to +4.0, p = 0.39).
- True new customers (signed up 2024 or later), **180-day window: 72.5% (158 of 218) vs 68.7% (373 of 543), +3.8 points (95% CI −3.3 to +10.9), p = 0.30, h = 0.08**; 90 days +1.4 (CI −5.8 to +8.5); 365 days 89.8% vs 88.3%. Completed first orders only, 180 days: 71.0% vs 68.4% (+2.6, CI −4.9 to +10.1).
- Spend in the year after the first order (new customers with 365 days of follow-up): $580 with the code vs $582 without (difference −$3, 95% CI −$119 to +$114; Mann-Whitney p = 0.54).
- With about 218 and 543 customers, the smallest difference detectable with 80% power is roughly 11 points: a modest real effect could be hidden.
- Cost: about $22,300 of discount on 1,139 orders over two years. First orders with the code are smaller (median $135 vs $153).

THE ANSWER THE DATA SUPPORT

No clear evidence that the code brings people back. Among genuinely new customers, 72.5% of code users and 68.7% of others ordered again within 180 days; the difference (+3.8 points) could be anywhere from 3 points lower to 11 higher. The apparent advantage in a naive "ever came back" count disappears once everyone gets the same follow-up window. Customers chose to use the code, so even a difference would not show the code caused it.

TRAPS

- **Time to return:** "ever came back" gives every customer a different follow-up; it produces the only significant number in this question, and it is an artifact of the measure.
- **First order in the data is not a first order** for customers who signed up before 2024.
- **Self-selection:** customers chose the code; "because" is not available without an experiment.
- **Statuses:** cancelled and returned first orders included or excluded without saying which.
- **Overlap with other codes:** 81 first orders used TRAIL15 or SUMMIT20 instead; they belong in the "no WELCOME10" group or are set aside, by a stated decision.

WHAT AN HONEST "NO CLEAR DIFFERENCE" SOUNDS LIKE HERE

"Within 180 days, 72.5% of new customers who used WELCOME10 ordered again, against 68.7% of new customers who did not: +3.8 points (95% CI −3.3 to +10.9; p = 0.30; h = 0.08). We found no clear difference; the data cannot rule

out an effect of up to about 11 points, and they do not support the claim that the code brings customers back."

FOLLOW-UP QUESTIONS TO ASK THE LEARNER

1. "What is your follow-up window, and how many customers did it exclude?"
2. "Whose first order is really a first order?"
3. "How would you design a fair test: offer the code to a random half of new signups for a month?"

WHAT A WEAK SUBMISSION TYPICALLY GETS WRONG

- Reports the naive +2.8 points ($p = 0.039$) as the answer.
- Compares repeat counts over different lengths of time.
- Says the code "causes" or "drives" loyalty.

ANSWER NOTES · QUESTION D: WHAT DID THE MARCH 2025 PRICE INCREASE DO TO SALES?

A SOUND ANALYSIS: TABLES AND STEPS

1. Merge `order_items.csv` with `orders.csv` (45,490 lines; every line matches) and identify repriced products from the price actually charged: each SKU has exactly one `unit_price` before 2025-03-01 and one after. Products whose charged price rose are the repriced group (133 of the 229 SKUs sold in both periods); the other 96 are the comparison group. Never use `products.price` for old orders.
2. Use completed orders; set aside the 12 bulk lines (25 to 39 units) and SKUs not sold in both periods (6 discontinued, 8 launched in late December 2025).
3. Make the comparison fair in time and against growth: units of repriced against unchanged products, March to December 2025 against March to December 2024 (a difference in differences). Check the same comparison before the change (January to February 2025 against 2024) as a placebo.
4. Get an interval: per-SKU log growth compared between the groups (Welch t), or a bootstrap over SKUs of the pooled ratio.
5. Look at the size of each rise (3.0% to 13.3%, median 5.7%) for a dose response, at category mix, and at revenue as well as units.

KEY NUMBERS (FROM THE INSTRUCTOR'S OWN RUN ON THE COURSE DATA)

- **The naive comparison (the trap):** repriced products sold 44.3 units a day in January to February 2025 and 50.8 a day in March to December: up 15% after a price rise. Unchanged products also rose (29.2 → 33.0). Seasonality and growth hide the drop.
- **The second trap, year over year without a comparison group:** repriced units March to December went 15,566 → 15,543 (−0.1%), which reads as "no effect". Unchanged products grew 8,611 → 10,086 (+17.1%) over the same

months: **difference in differences −14.8%**. The same calculation before the change (January to February) gives +3.9%: the two groups moved together until March.

- Per SKU (219 SKUs with at least 5 units in each window): repriced units grew ×1.006, unchanged ×1.152; **ratio ×0.874 (95% CI ×0.795 to ×0.960), Welch p = 0.005, d = −0.39** (Mann-Whitney p = 0.0004). Pooled over SKUs: **−14.2% (95% bootstrap CI −19.2% to −8.7%)**.
- Dose response: larger rises did not lose more (Spearman 0.14, p = 0.11): the drop is about the same whatever the size of the rise, so the data do not support a per-percent elasticity. Within categories the difference is negative in 7 of 9 (−26% Sleeping to +9% Climbing; Tents +1%): small groups, noisy, but mostly the same sign.
- Revenue on repriced products, March to December: $819,994 → $880,294 (+7.4%), against +20.7% for unchanged products: the higher price did not make up for the lost units, relative to trend.
- Valuing January to February 2025 orders at today's `products.price` overstates their revenue by 3.2%; 61.4% of lines before the change were charged a different price from today's list price.

THE ANSWER THE DATA SUPPORT

A clear drop in volume. Compared with products whose price did not change, over the same months a year apart, repriced products sold about 14% fewer units than their trend (95% bootstrap CI about 9% to 19% fewer; per-SKU ratio ×0.87, CI ×0.80 to ×0.96). Before the change the two groups moved together (+3.9%), so the gap opens at the price change. Revenue from the repriced products still grew (+7.4%), but far less than for unchanged products (+20.7%). Pinecrest chose which products to reprice, so this is a well-supported estimate of what happened, not a randomized test.

TRAPS

- **Before and after across seasons:** January and February are the slowest months, so the repriced products look 15% busier afterward.
- **Growth:** the whole company grew in 2025; repriced units flat year over year looks like "no effect" until the unchanged products (+17%) are put beside them.
- **Wrong price column:** `products.price` is today's price; repriced products must be found from `order_items.unit_price`.
- **Merge that multiplies rows:** summing `order_total` after merging lines (Common Mistake 2) inflates revenue.
- **Bulk lines and launches:** 12 bulk lines and late launches distort SKU-level ratios.
- **Selection:** Pinecrest chose which products to reprice; the groups may have been on different paths anyway (the placebo check speaks to this, and it is clean).

WHAT A SOUND CLAIM, AND AN OVER-CLAIM, SOUND LIKE HERE

Sound: "Repriced products sold about 14% fewer units than comparable unchanged products, relative to the same months a year earlier (95% CI 9% to 19% fewer; per-SKU p = 0.005). Before March the two groups moved together, so the drop starts with the price change. The products were not chosen at random." Over-claims to mark down: "the

price increase cut Pinecrest's sales by 14%" (it is the repriced products' units, against their trend, not company sales or revenue); "every 1% of price costs about 2% of volume" (no dose response in these data); "customers will not accept higher prices, so roll them back" (a decision the analysis does not make: revenue on those products still grew); a −14.8% with no interval or placebo.

FOLLOW-UP QUESTIONS TO ASK THE LEARNER

1. "How did you decide which products were repriced, and from which column?"
2. "What happened to the unchanged products over the same months, and before March?"
3. "Did the bigger price rises lose more volume? What does that mean for a sentence about elasticity?"
4. "Is your result about units or revenue, and which one would leadership use to decide?"

WHAT A WEAK SUBMISSION TYPICALLY GETS WRONG

- Before-and-after on repriced products only: "sales went up 15% after the price rise".
- Year over year on repriced products only: "units held steady, so the increase cost nothing".
- Uses `products.price` to identify or to value the old orders.
- Reports the drop as a cut in total company sales, or as a price elasticity.

HOW THESE NOTES WERE MADE

Every number above was produced by the kit author's own analysis of the shipped course data (pandas 3.0, SciPy, statsmodels; seeded bootstraps), following the steps listed for each question. Learners may reach slightly different numbers with other reasonable definitions (another follow-up window, means instead of medians, a different treatment of the bulk lines): judge the definitions and the reasoning, not agreement to the cent.

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